

Practice Set

BCACC601

Advanced Database Management System with PL/SQL

Module-2

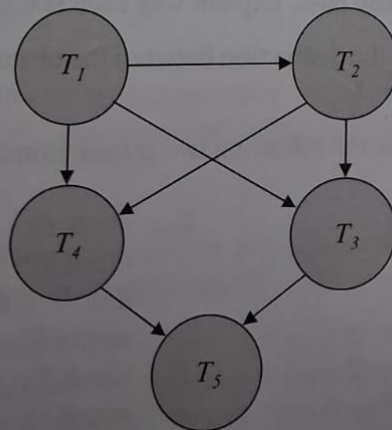
1. What is Transaction?
2. Give an example of Transaction.
3. What is ACID property?
4. What are different transaction states?
5. State the difference between failure and abort state of transaction.
6. What are the benefits of concurrent execution of transactions?
7. What is serializability?
8. What is view serializability?
9. What is conflict serializability?
10. How to test conflict serializability?
11. What is deadlock?
12. State one difference between deadlock and starvation.
13. State 2 deadlock prevention strategies.
14. State wait-die scheme as deadlock prevention strategy.
15. State wound-wait scheme as deadlock prevention strategy.
16. What is the difference between commit and rollback?
17. What is the necessity of locking?
18. State 2 modes in which data items can be locked.
19. State basic principle of 2 phase locking.
20. What is growing phase in 2 phase locking?
21. What is shrinking phase in 2 phase locking?
22. What is lock manager?
23. State basic principle of timestamp based locking protocol.
24. What is Multiple Granularity (MG)?
25. What is fine granularity?
26. What is coarse granularity?
27. State 2 differences between fine granularity and coarse granularity.
28. What is the effect of time stamp on serializability?
29. What is W-timestamp?
30. What is R-timestamp?
31. State basic principle of recovery algorithm.
32. What is Transaction Failure?

33. What is the role of data storage in recovery?
34. What is the role of grant and revoke in database?
35. State the necessity of shadow copy in terms of recovery.
36. What is immediate database modification scheme?
37. What is deferred database modification scheme?
38. What is role of log file in recovery?
39. State the necessity of checkpoints in database.
40. State the differences between logical errors and system errors.

1. Describe ACID property in terms of Transactions in database.
2. Explain transaction states with diagrams.
3. What is scheduling? Give examples of different scheduling schemes.
4. How does view serializability is tested?
5. What are the ways to test conflict serializability?
6. Illustrate 2 phase locking protocol with examples and diagrams.
7. Illustrate timestamp based locking protocol with examples and diagrams.
8. State the necessity of locking in terms of concurrency control in database.
9. Explain Multiple Granularity concept with example.
10. What is the importance of Multiple Granularity in terms of concurrency control?
11. State the differences between fine granularity and coarse granularity in detail.
12. What are the necessity of lock table and lock manager in concurrency control?
13. Explain different deadlock prevention strategies.
14. State the differences between wait-die and wound-wait deadlock prevention strategies.
15. What is the role of timestamp order in maintaining serializability?
16. Write a short notes on 2 phase locking.
17. Write a short notes on timestamp based locking.
18. Write a short notes on Multiple Granularity (MG).
19. Classify different types of failure in terms of database recovery.
20. Explain log-based recovery protocol in detail.
21. Illustrate the role of checkpoints in database recovery.

Practice Exercises

- ✓ 17.1 Suppose that there is a database system that never fails. Is a recovery manager required for this system? (5)
- 17.2 Consider a file system such as the one on your favorite operating system.
- What are the steps involved in the creation and deletion of files and in writing data to a file?
 - Explain how the issues of atomicity and durability are relevant to the creation and deletion of files and to writing data to files.
- ✓ 17.3 Database-system implementers have paid much more attention to the ACID properties than have file-system implementers. Why might this be the case? (5)
- ✓ 17.4 What class or classes of storage can be used to ensure durability? Why? (5)
- ✓ 17.5 Since every conflict-serializable schedule is view serializable, why do we emphasize conflict serializability rather than view serializability? (5)
- ✓ 17.6 Consider the precedence graph of Figure 17.16. Is the corresponding schedule conflict serializable? Explain your answer. (5)
- ✓ 17.7 What is a cascadeless schedule? Why is cascadelessness of schedules desirable? Are there any circumstances under which it would be desirable to allow noncascadeless schedules? Explain your answer. (5)
- 17.8 The **lost update** anomaly is said to occur if a transaction T_j reads a data item, then another transaction T_k writes the data item (possibly based on a previous read), after which T_j writes the data item. The update performed by T_k has been lost, since the update done by T_j ignored the value written by T_k .



- a. Give an example of a schedule showing the lost update anomaly.
 - b. Give an example schedule to show that the lost update anomaly is possible with the **read committed** isolation level.
 - c. Explain why the lost update anomaly is not possible with the **repeatable read** isolation level.
- 17.9 Consider a database for a bank where the database system uses snapshot isolation. Describe a particular scenario in which a nonserializable execution occurs that would present a problem for the bank.
- 17.10 Consider a database for an airline where the database system uses snapshot isolation. Describe a particular scenario in which a nonserializable execution occurs, but the airline may be willing to accept it in order to gain better overall performance.
- 17.11 The definition of a schedule assumes that operations can be totally ordered by time. Consider a database system that runs on a system with multiple processors, where it is not always possible to establish an exact ordering between operations that executed on different processors. However, operations on a data item can be totally ordered.
- Does this situation cause any problem for the definition of conflict serializability? Explain your answer.

Exercises

- ✓ 17.12 List the ACID properties. Explain the usefulness of each. (5)
- 17.13 During its execution, a transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur.
- ✓ 17.14 Explain the distinction between the terms *serial schedule* and *serializable schedule*. (5)
- 17.15 Consider the following two transactions:

```

T13: read(A);
      read(B);
      if A = 0 then B := B + 1;
      write(B).
T14: read(B);
      read(A);
      if B = 0 then A := A + 1;
  
```

Practice Exercises

- ✓ 18.1 Show that the two-phase locking protocol ensures conflict serializability and that transactions can be serialized according to their lock points. (5)
- ✓ 18.2 Consider the following two transactions: (5)

T_{34} : read(A);
 read(B);
 if $A = 0$ then $B := B + 1$;
 write(B).

T_{35} : read(B);
 read(A);
 if $B = 0$ then $A := A + 1$;
 write(A).

Add lock and unlock instructions to transactions T_{31} and T_{32} so that they observe the two-phase locking protocol. Can the execution of these transactions result in a deadlock?

- ✓ 18.3 What benefit does rigorous two-phase locking provide? How does it compare with other forms of two-phase locking? (5)
- 18.4 Consider a database organized in the form of a rooted tree. Suppose that we insert a dummy vertex between each pair of vertices. Show that, if we follow the tree protocol on the new tree, we get better concurrency than if we follow the tree protocol on the original tree.
- 18.5 Show by example that there are schedules possible under the tree protocol that are not possible under the two-phase locking protocol, and vice versa.
- 18.6 Locking is not done explicitly in persistent programming languages. Rather, objects (or the corresponding pages) must be locked when the objects are accessed. Most modern operating systems allow the user to set access protections (no access, read, write) on pages, and memory access that violate the access protections result in a protection violation (see the Unix `mprotect` command, for example). Describe how the access-protection mechanism can be used for page-level locking in a persistent programming language.
- 18.7 Consider a database system that includes an atomic **increment** operation, in addition to the read and write operations. Let V be the value of data item X . The operation

increment(X) by C

- ✓ 19.14 Explain the difference between the three storage types—volatile, nonvolatile, and stable—in terms of I/O cost. (5)
- ✓ 19.15 Stable storage cannot be implemented. (5)
- Explain why it cannot be.
 - Explain how database systems deal with this problem.
- ✓ 19.16 Explain how the database may become inconsistent if some log records pertaining to a block are not output to stable storage before the block is output to disk. (5)
- 19.17 Outline the drawbacks of the no-steal and force buffer management policies.
- 19.18 Suppose two-phase locking is used, but exclusive locks are released early, that is, locking is not done in a strict two-phase manner. Give an example to show why transaction rollback can result in a wrong final state, when using the log-based recovery algorithm.
- 19.19 Physiological redo logging can reduce logging overheads significantly, especially with a slotted page record organization. Explain why.
- 19.20 Explain why logical undo logging is used widely, whereas logical redo logging (other than physiological redo logging) is rarely used.
- 19.21 Consider the log in Figure 19.5. Suppose there is a crash just before the log record $\langle T_0 \text{ abort} \rangle$ is written out. Explain what would happen during recovery.
- 19.22 Suppose there is a transaction that has been running for a very long time but has performed very few updates.
- What effect would the transaction have on recovery time with the recovery algorithm of Section 19.4, and with the ARIES recovery algorithm?
 - What effect would the transaction have on deletion of old log records?
- 19.23 Consider the log in Figure 19.7. Suppose there is a crash during recovery, just before the operation abort log record is written for operation O_1 . Explain what will happen when the system recovers again.
- 19.24 Compare log-based recovery with the shadow-copy scheme in terms of their overheads for the case when data are being added to newly allocated disk pages (in other words, there is no old value to be restored in case the transaction aborts).
- 19.25 In the ARIES recovery algorithm:

1. Define Transaction with example. Explain different states of transactions with examples and diagrams.
2. Give examples of different scheduling schemes with their advantages and disadvantages.
3. State the differences between conflict serializability and view serializability with examples.
4. Demonstrate the role of precedence graph in terms of conflict serializability testing.
5. Write short notes on – i. Serializability ii. Scheduling
6. Describe the advantages and disadvantages of 2 phase locking protocol in detail.
7. Describe growth phase and shrinking phase of 2 phase locking protocol with diagrams.
8. Describe the advantages and disadvantages of timestamp based locking protocol in detail.
9. Illustrate the role of lock table and lock manager in concurrency control with diagrams.
10. State the importance of W-timestamp and R-timestamp while designing timestamp ordering protocol.
11. Explain the necessity of locking in terms of real time transactions.
12. Justify - Timestamp protocol ensures freedom from deadlock as no transaction ever waits.
13. Write short notes on – i. Two Phase Locking ii. Timestamp based Locking
14. State the differences between strict two phase locking and rigorous two phase locking protocol with examples.
15. Explain the role of the following in terms of database transactions – commit, rollback, grant, revoke.
16. Demonstrate in detail the advantages and disadvantages of wait-die and wound-wait deadlock prevention strategies.
17. Write a database recovery algorithm with proper steps and example.
18. Explain the role of volatile storage and non-volatile storage in terms of database recovery.
19. What is log based recovery? Explain the role of log file in data recovery with example.
20. Write short notes on – i. Checkpoints ii. Log Based Recovery